15

The Political Economy of Transitions

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15.1 Introduction: The Political Economies of Transition

In the study of socio-technical transitions and of pathways to sustainability more broadly, there has been increasing attention to questions of politics and governance (Meadowcroft 2009; Scoones et al. 2015). Studies have highlighted, for example, the important role of institutions, ideas and political interests in shaping transition pathways (Kern 2011). There is also growing interest in applying theoretical insights from different approaches to political economy as discussed in Section 15.2. However, significant scope remains to strengthen accounts which centre relations of power in seeking to understand who wins, who loses, how and why from prevailing governance arrangements and social-technical configurations (Newell 2018). One of the benefits of political economy research on transitions is that it relates the underlying causes of problems that transitions are said to address to the structures within which they will be governed (Newell 2018). This can lead to an overly pessimistic account of why change does not occur because of its focus on how incumbent or hegemonic power gets reproduced by dominant social and economic actors charged with addressing the problems they helped to create. On the other hand, it provides invaluable resources for critically assessing current landscapes of power in order to then explore the prospects of change and from whence openings for change might come. Concretely, this more practical application of political economy analysis can take the form of analysis of the potential for coalitions and broader alliances for change, for example (Meckling et al. 2015; Hess 2018, 2019).

Classic political economists such as Adam Smith, David Ricardo and John Maynard Keynes sought to understand the processes of wealth creation and exchange in society, while critical political economists, most notably Karl Marx, explored the injustices and exploitation that these relations rely upon to generate wealth for the few at the expense of the many. It is important to note, firstly, that there are many political economies and ways of understanding power, therefore,

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production and reproduction of order in society. The starting point is often the relationship between states and markets (Strange 1988) and the different forms that this relationship can take more liberal, coordinated and regulated, for example (Hall and Soskice 2001): the different ways in which they are governed and their distributional impacts. Political economy accounts tend to place centrally control over key areas of the economy from production and trade, to finance and technology as a means of understanding the material basis of uneven distributions of power. This provides a basis for understanding the ways in which governance arrangements seek simultaneously to expand the economy and manage the social and ecological conflicts which arise from that expansion. We will suggest here that political economy approaches can be refined and adapted to shed important light on the ways in which dominant socio-technical configurations reflect broader social and economic relations in society, just as disruptions associated with transitions can shift the power relations that political economy accounts seek to explain.

15.2 Historical and Thematic Development: From Governance and Politics to Political Economy

Early critiques of the neglect of questions of governance, politics and power in transition studies (Meadowcroft 2009; Shove and Walker 2007), led to leading authors in the field attempting to 'bring politics in' (Geels 2014, 2019). Transition scholars have incorporated insights from political science in a number of different ways. Some scholars have developed historical accounts of key institutions involved in the governance of transitions (Lockwood et al. 2016) or focused on 'feedback effects' (Lockwood 2015): the generation of benefits from transitions for key actors that help to secure their support for them and the role of 'political coalitions' in supporting and resisting change (Hess 2018). Other research has focused on questions of incumbency and elite power (Sovacool and Brisbois 2019) which is often exercised to resist or slow transitions threatening to status quo interests. This is sometimes organised around particular regimes of finance and production (Baker et al. 2014) and increasingly tied to questions of hegemony (Ford and Newell 2021a). In turn, this has solicited growing interest in the role and nature of the state: not just its 'entrepreneurial' role in supporting innovation (Mazzucato 2011), but also adopting a relational view of tensions and complexities which arise from the multiple roles that states are expected to play in relation to transitions (Silvester & Fisker 2023; Johnstone and Newell 2018). Rather than adopt a monolithic view of the state, scholarship on varieties of state-market relations and forms of capitalism has explored the different approaches of coordinated (more social democratic) versus liberal market economies in how and by whom transitions are organised (Ćetković & Buzogány 2016).

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Alongside growing attention to governance and the state, there has been growing interest in the global governance of key issue areas such as energy, water and food; systems that are of interest to transition scholars (Van de Graaf 2013; Goldthau and Witte 2010), though the global governance and global political economy (GPE) of transitions per se remains a neglected area (Newell 2020, 2021). In terms of the conceptualisation of the global governance of transitions, critical GPE helps to inform an understanding of 'the broader political and economic landscapes which shape transition pathways, the global interrelationships between national level transitions and to an appreciation of the shifting role of the state in a context of globalisation'. This helps' not just to account for existing distributions of power and wealth in society, but to explore and engage with the potential for transformation beyond existing systems' (Newell 2020a: 344). In particular, such accounts can help (i) situate sustainability transitions within particular historical conjunctures, thereby contributing to work on 'deep transitions'(Kanger and Schot, 2018). This body of work locates a series of connected and sustained fundamental transformations of a wide range of socio-technical systems in a similar direction such as moves towards increased labour productivity, mechanisation, reliance on fossil fuels, resource-intensity, energy-intensity and reliance on global value chains. A GPE account would relate these transformations to shifts in the stages of capitalism from Fordist models of production, for example, to a more financialised stage of late capitalism. Such an account can also provide rich historical accounts of technological, social and political lock in over time (Malm 2016): how incumbent pathways are often supported and reinforced while alternatives are marginalised (Unruh 2000) (ii) provide an account of the inter-relationships between transitions in different parts of the world. These reflect the uneven distribution of power in the international system and the ways in which both benefits are captured by wealthier groups and costs passed on to the poorer ones through spatial and temporal fixes (Newell 2021a,b) (iii) appreciate how the state's insertion into the GPE shapes the degree of policy space and autonomy that states have to chart their own transition pathways. This space is affected by the nature of the trade, investment and aid relationships they have with other states and regions.

Going further still in this direction, there is increasing attention in political economy accounts to capitalism (Feola 2020) and extractivism (Gudynas 2021); the ways in which transitions are organised (and on whose behalf) in capitalist economies (Newell and Phillips 2016) and scope for transitions within and beyond capitalism (Newell and Paterson 2010) including deeper transformations. Given the emphasis on distributional questions in political economy, it is unsurprising that there has been increasing attention to questions of justice which inevitably foreground questions of power, politics and social relations. These have been applied to questions of just transition (Swilling and Annecke 2012; Newell and

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Mulvaney 2013) in relation to specific sectors such as energy (Healy et al. 2019), but also explored in relation to the role of global governance institutions in frustrating and enabling just transitions (Newell et al. 2023).

15.3 Empirical Application: The Political Economy of Energy Transitions in India

This section provides an empirical illustration of the political economy of energy transitions using India as a case study of a country that operates as a globally significant 'rising power' shaping energy transitions elsewhere in the world (Power et al. 2016), while also embroiled in its own national and state level transition politics.

15.3.1 The Indian State's Material Lock-in to Coal

This sub-section discusses the various fossil fuel lock-ins that underpin India's energy sector, focusing primarily on the material interests of the Indian state. The Indian state has historically played a disproportionately large role in India's energy sector through its monopoly over coal resources, its direct ownership of the majority of thermal power plants, and as the primary source of credit for energy infrastructure through publicly owned banks. State-controlled actors such as Coal India Limited and NTPC (formerly National Thermal Power Corporation) have dominated India's energy system. These quasi-state entities have played an important developmental role, spending significant parts of their revenues on local health and educational services in the areas where they operate, thereby securing their social licence and creating a social lock-in for their largesse. Enterprises such as Coal India have become deeply embedded in India's coal districts in multiple ways by creating and maintaining infrastructure, creating jobs and contributing key revenues crucial for the operation of local governments (Chandra 2018).

The Indian state has a substantial economic interest in fossil fuel infrastructures at the union and regional levels, in at least three ways. Firstly, state-owned fossil enterprises contribute significantly to public finances through taxes, duties and royalties. One estimate suggests that the union government depends on coal, petroleum and gas for ~25% of its total revenue receipts (Gambhir et al. 2021). Secondly, through the direct and indirect ownership of fossil assets, state-owned banks have substantially financed India's thermal fleet. The average age of India's thermal fleet is just 12 years. Hence, according to some models, a rapid transition consistent with a well-below 2°C scenario could result in capacity stranding of the order of 133–237 GW (Malik et al. 2020), leaving powerful asset

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owners and their ultimate backers – publicly owned banks – exposed. Thirdly, coal revenues effectively cross-subsidise passenger railway fares, accounting for approximately 44% of Indian Railways' freight revenues in 2017 (Kamboj and Tongia 2018).

Beyond these clear dependencies, there are also more obfuscated ways in which the Indian state colludes with private capital as part of the coal regime. For instance, state actors in coastal regions are building new port infrastructure to make viable imported coal-fired power plants, thus benefiting a nexus of interests including coal exporting nations and some of India's largest conglomerates (Oskarsson et al. 2021). Most existing literature typically elides over the most direct forms of collusion, but links between the rights to privately mine coal and political funding have been alleged in the media. Indeed, India's union government in 2014 was brought down by allegations of corruption in the discretionary allocation of coal blocks. Some scholars have argued that these dependencies and entanglements are deepening, that India's lock-in to coal is intensifying over time (Roy and Schaffartzik 2021).

But the material interests of the Indian state only partly account for the resilience of fossil fuels in India. Fossil energy is also discursively and ideologically intertwined with the project of India's development and state-building, a phenomenon that has been variously labelled as 'resource nationalism' (Lahiri-Dutt 2016) or 'fossil developmentalism' (Chatterjee 2020). An estimated 3.6 million people are directly or indirectly employed in coal mining and thermal power generation, with an additional half-a-million coal pensioners (Pai 2021). In addition, an illegal economy and criminality have developed around pilfering and theft of coal, sustained by the complicity of local political elites the (Lecavalier and Harrington 2017). The socio-politics of labour in coal belts are also substantially co-determined by gendered and caste-based oppressions (Nayak 2022). Economic and social forces resulting from these complex entanglements are strongly reflected in the electoral politics of coal-producing regions, and in India's stance against a full coal phase-out on the international stage.

15.3.2 Governance Lock-ins Related to Indian Federalism and Electricity Governance

Despite progress towards clean energy supply in recent years, roughly three-quarters of India's electricity in 2022 was generated from coal-fuelled thermal power plants (Bhatia 2023). Thus, the governance of India's energy transition is intricately tied to the governance of its electricity sector. The Constitution prescribes that the electricity sector's governance be shared between the national and subnational governments, with the former in a direction-setting and the latter in an operational role.

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However, there is a misalignment of incentives between these levels regarding how to govern the energy transition.

First, the union and state governments have different goals in governing the power sector. In its direction-setting role, the union government espouses greater ambition towards decarbonisation than sub-national governments, who tend to resist top-down renewable energy (RE) targets and instead, prefer to govern in a manner that secures low-cost power for their constituents and the stability of a financially and technically strained system.

Second, not all sub-national governments will benefit equally from the transition. Given the different spatial distribution of coal and renewable resources, the transition could exacerbate inter-regional inequities within India's federal polity (Newell et al. 2011): the potentially adverse impacts of transitions – loss of employment and energy-intensive industries – are likely to be strongly concentrated in less-wealthy coal mining states in Eastern India, whereas green employment is likely to be dispersed across richer Western and Central states (Ordonez et al. 2023). Unless the governance challenge of equitably distributing the gains from a just transition is addressed, pro-transition forces are liable to be trumped by concerns for federal cohesion.

Third, the transition risks upending the existing political economy of electricity distribution, which rests on a delicately balanced form of welfarism. Electricity tariffs are highly politically sensitive. India's welfarist historical institutions have bred a 'cross-subsidisation' model whereby electricity tariffs are set in a manner such that industrial and commercial consumers overpay to enable subsidised consumption by the poorest residential and agricultural consumers. Interventionist sub-national governments have found ways to influence ostensibly independent electricity regulators – most often by appointing politically aligned members – to ensure tariffs remain artificially low for key constituencies, and these are carefully managed by sub-national governments (Dubash et al., 2018). The combination of decentralised solar and storage provides a pathway to upend this delicate balance by allowing high-paying customers to 'migrate' away from the grid, potentially causing a breakdown in the business model for a major welfare service (Dubash et al., 2019) and its underpinning social contract. Publicly owned utilities face the downside risks of the breakdown of this business model. The Indian state is thus implicated in ensuring that the energy transition is orderly and the utilities are protected, even if that implies a slower transition.

15.3.3 Winners and Losers in the Renewable Energy Sector

Political economic forces are as instrumental in shaping the contours of India's emerging green energy system as the fossil system. In line with India's broader

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move towards greater liberalisation and privatisation, its RE capacity is predominantly owned and operated by private rather than governmental enterprises. Latest cost and ownership trends suggest that most future greenfield capacity will be privately owned solar and wind projects. These politics have also fostered scalar biases towards mega-scale 'solar parks' over distributed energy in India's RE rollout, which could enable the capture of the benefits of the energy transition by well-capitalised energy-asset-owning firms and elites (Sharma and Bhatia 2022). Simultaneously, large-scale RE projects have been found to unjustly allocate most costs to the most vulnerable, for instance by enclosing common lands or 'wastelands', which disproportionately affects pastoral communities (Yenneti and Day 2016). Thus, moves towards diversification and decarbonisation of the energy mix may well fall short of a just transition.

Furthermore, the politics of green energy are shaped by the politics of trade and national competitiveness. While India's energy policy framework has de jure promoted both deployment and manufacturing, in practice it has successfully driven RE deployment without developing a thriving green manufacturing base. Key dynamics driving this outcome are illustrated by a battle between a transnational, deployment-focused coalition and a domestic, manufacturing-focused coalition. As a late industrialiser, India sought to support its budding solar manufacturers through a domestic content requirements clause in its flagship National Solar Mission. A challenge by the US at the World Trade Organisation (WTO) forced India to retreat from these policies, slowing the momentum of solar manufacturers and driving them to seek additional and alternate protections (Tagotra 2017). However, a coalition of solar PV exporting nations, their lead manufacturing firms, and domestic project developers seeking to deliver low-cost power using imported panels emerged as a strong countervailing force (Behuria 2020). The latter's success in influencing policy has contributed to India's RE capacity additions far outpacing its green manufacturing capacities.

Post-COVID, the prospect of India's energy security becoming dependent on technology imports has induced a renewed policy push for self-reliance ('aatmanirbharta') through manufacturing, which will likely become a central part of the political economy of India's energy transition in the coming years. It is not yet clear who would benefit from this shift and how, but the initial signs suggest that the Indian state is poised to direct substantial fiscal resources towards the large energy-industrial corporations. It has put in place a spate of production-linked incentive (PLI) policies[1] aimed at stimulating high-tech manufacturing segments across 14 key sectors, which are accessible primarily to established energy-industrial incumbents (dubbed 'national manufacturing champions'). This also

[1] www.investindia.gov.in/production-linked-incentives-schemes-india

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suggests that India's energy transition could be tilted towards large energy and industrial incumbents rather than towards disruptive green energy players emerging from protected niches.

15.4 Ongoing Debates & Further Needs on the Political Economy of Transitions

Firstly, we can see ongoing debates in transition studies about how best to understand the power relations which most scholars now accept are critical to understand and engage with as both an academic enterprise and in order to effect real-world change. Different approaches emphasise discursive power, institutional power as well as more material expressions of power (Isoaho & Karhunmaa, 2019; Kuzemko et al. 2016). In reality these are not mutually exclusive and often overlap and combine to both enable and frustrate different transition pathways. Despite recent advances, we still lack a deeper and more holistic understanding of the state and how its different dimensions of power and arenas of function interrelate and the implications of this for transitions. This includes, across different state agencies and levels of sub-national decision-making as the Indian case makes clear. But also how military, welfare, foreign policy and entrepreneurial functions of the state impact the pursuit and nature of sustainability transitions. Related to this is the need for a more nuanced understanding of industry in (Newell 2025) the 'business' of transition (Newell 2020a). Not just the lobbying and financial and political ties which bind the state and capital in different ways across diverse governance systems, but also the everyday politics of industrial organisation and re-ordering along complex power-laden value chains and through global production networks which shape transition pathways (Baker and Sovacool 2017). It is here that opportunities for challenging incumbent power and de-stabilising dominant regimes can become more visible (Leipprand & Flachsland 2018).

Secondly, questions of justice are central to any political economy account of transition processes. These centre on the politics of who is included and excluded from decision-making over pathways to sustainability and what this means in terms of the winners and losers from transitions and deeper transformations. Scholars and practitioners have started to analyse the inevitable trade-offs and tensions between procedural and distributional dimensions (Ciplet and Harrison 2020) including debates about how to reconcile speed and inclusivity (Roberts et al. 2018; Newell and Simms 2020; Sovacool 2016; Kumar et al. 2021). For example, does the 'need for speed' mean that slower and more inclusive forms of deliberation are not viable or is deepening democracy a prerequisite to more progressive transformations? (Stirling 2014). Political economy accounts that emphasise the ways in which the state and dominant systems of governance both

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reflect and are embedded in broader social relations of class, caste, race, gender and coloniality help to broaden and deepen the analytical frame of transitions research (Lennon 2017; Newell 2021a). Attention to coloniality in particular underscores the need to attend to restorative justice in transition debates, something indigenous communities know only too well (Gilio-Whitaker 2019). But such accounts need to be grounded in concrete struggles for just transitions which seek to challenge and overcome the multiple hierarchies, gendered and racialised assumptions and biases which structure and delimit the scope for more inclusive and progressive transitions (Bell et al. 2020)

Thirdly, a key challenge is how to conceptually connect global, national and local political economies as our analysis above shows they must. Global dynamics still need to be located in specific political economies to make sense of the peculiarities and situated nature of national and sectoral transitions (Baker et al. 2014) and to draw on 'local' research traditions and trajectories to make sense of them (Broto et al. 2018). The nature of this global terrain is, of course, shifting amid major realignments in geopolitics with huge implications for the prospects of energy transitions. The pace of transition away from gas, for example, has been sharply affected by Russia's invasion of Ukraine sparking conflicting demands: both to exploit more fossil fuels locally and to chart a path away from fossil fuels to avoid such dependencies in future. Refining accounts of the geopolitics of transitions building on existing strands of work (Shen and Power 2017; Power et al. 2016; Blondeel et al. 2021) will clearly be increasingly important in transitions research.

Fourthly, intersections with ecology and more than human transitions means broadening and challenging political economy by bringing in insights from political ecology (Lawhon and Murphy 2012) to understand the 'ecologies of transition' (Newell 2021a): how demands for water, energy and land combine and solutions to one problem often exacerbate others. In turn, this requires an appreciation of how transitions 'travel' as costs and benefits are allocated unevenly between and within societies along temporal and spatial lines. Concrete contemporary examples might be how transitions in transport towards the use of biofuels produce conflicts over land and food security elsewhere in the world (Smith 2000) or how the drive to electrification in richer parts of the world intensifies struggles over mining for critical minerals in places like the 'lithium triangle' in Latin America (Hernandez and Newell 2022). Normatively, this gives rise to strategic questions of how to challenge dominant modes of extractivism and the political economies that give rise to them.

Finally, though we started the chapter by emphasising how political economy accounts of transitions are often grounded in analysis of economic structures, there is growing attention to cultural political economy which draws attention to norms, behaviours and practices in the production of energy and food cultures, for example

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(Strauss et al. 2013), such as work on petro-masculinities (Daggett 2018). Political economy accounts which are more attentive to the power of discourse, ideology, knowledge politics and the everyday cultural reproduction of patterns of production and consumption through social practice (Shove et al. 2012) will be better placed to capture the mutually reinforcing nature of material, institutional, discursive and social and cultural power as they are expressed in sustainability transitions.

As power relations reconfigure and the political systems within which transitions are organised and contested evolve in relation to shifting 'landscape' pressures, the diverse set of tools and approaches introduced in this chapter provide a rich set of conceptual and empirical resources to draw on in making sense of contemporary sustainability transitions.

References

Baker, L., and B. Sovacool (2017) 'The Political Economy of Technological Capabilities and Global Production Networks in South Africa's Wind and Solar Photovoltaic (PV) Industries'. *Political Geography* 60: 1–12.

Baker, L., P. Newell, and J. Phillips (2014) 'The Political Economy of Energy Transitions: The Case of South Africa'. *New Political Economy* 19: 791–818.

Behuria, P. (2020) 'The Politics of Late Late Development in Renewable Energy Sectors: Dependency and Contradictory Tensions in India's National Solar Mission'. *World Development* 126 (February): 104726. https://doi.org/10.1016/j.worlddev.2019.104726.

Bell, S. E., C. Daggett, and C. Labuski (2020) 'Toward Feminist Energy Systems: Why Adding Women and Solar Panels is not Enough'. *Energy Research & Social Science* 68: 101557.

Bhatia, Parth (2023) 'India's State-Led Electricity Transition: A Review of Techno-Economic, Socio-Technical and Political Perspectives'. *Energy Research & Social Science* 102 (August): 103184. https://doi.org/10.1016/j.erss.2023.103184.

Blondeel, M., M. J. Bradshaw, G. Bridge, and C. Kuzemko (2021) 'The Geopolitics of Energy System Transformation: A Review'. *Geography Compass* 15(7): e12580.

Broto, V. C., I. Baptista, J. Kirshner, S. Smith, and S. N. Alves (2018) 'Energy Justice and Sustainability Transitions in Mozambique'. *Applied Energy* 228: 645–655.

Ćetković, S., and A. Buzogány (2016) 'Varieties of Capitalism and Clean Energy Transitions in the European Union: When Renewable Energy Hits Different Logics'. *Climate Policy* 16(5): 642–657.

Chandra, R. (2018) 'Adaptive State Capitalism in the Indian Coal Industry'. PhD Thesis, Boston, MA: Harvard University. https://dash.harvard.edu/handle/1/41127494.

Chatterjee, E. (2020) 'The Asian Anthropocene: Electricity and Fossil Developmentalism'. *The Journal of Asian Studies* 79(1): 3–24. https://doi.org/10.1017/S0021911819000573.

Ciplet, D., and J. L. Harrison (2020) 'Transition Tensions: Mapping Conflicts in Movements for a Just and Sustainable Transition'. *Environmental Politics* 29(3): 435–456.

Daggett, C. (2018) 'Petro-masculinity: Fossil Fuels and Authoritarian Desire'. *Millennium* 47(1): 25–44.

Dubash, N. K., A. K. Swain, and P. Bhatia (2019) 'The Disruptive Politics of Renewable Energy'. *The India Forum* May. www.theindiaforum.in/article/disruptive-politics-renewable-energy.

Dubash, N. K., S. S. Kale, and R. Bharvirkar, (eds). (2018) *Mapping Power: The Political Economy of Electricity in India's States*. New Delhi: Oxford University Press.

Feola, G. (2020) 'Capitalism in Sustainability Transitions Research: Time for a Critical Turn?' *Environmental Innovation and Societal Transitions*, 35: 241–250.

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Gambhir, A., A. Sreenivas, and A. Ketkar (2021) 'Energy: Taxes and Transition in India'. Working Paper. Pune: Prayas (Energy Group). www.prayaspune.org/peg/publications/item/485-energy-taxes-and-transition-in-india.html.
Geels, F. (2014) 'Regime Resistance against Low-Carbon Transitions: Introducing Politics and Power into the Multi-level Perspectives'. *Theory, Culture & Society* 31(5): 21–40.
Geels, F. (2019) 'Socio-technical Transitions to Sustainability: A Review of Criticisms and Elaborations of the Multi-Level Perspective'. *Current Opinion in Environmental Sustainability* 39: 187–201.
Gilio-Whitaker, D. (2019) *As Long as Grass Grows: The Indigenous Fight for Environmental Justice, from Colonization to Standing Rock*. Beacon Press.
Goldthau, A., and J. M. Witte (eds.) (2010) *Global Energy Governance: The New Rules of the Game*. Washington, DC: Brookings Press.
Gudynas, E. (2021) *Extractivisms: Politics, Economy and Ecology*. Rugby, UK: Practical Action Publishing.
Hall, P., and D. Soskice (2001) *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage*. Oxford: Oxford University Press.
Healy, N., J. C. Stephens, and S. A. Malin (2019) 'Embodied Energy Injustices: Unveiling and Politicizing the Transboundary Harms of Fossil Fuel Extractivism and Fossil Fuel Supply Chains'. *Energy Research & Social Science* 48: 219–234.
Hernandez, D. S., and P. Newell (2022) 'Oro Blanco: Assembling Extractivism in the Lithium Triangle'. *The Journal of Peasant Studies* 49(5): 945–968.
Hess, D. J. (2018) 'Sustainability Transitions: A Political Coalition Perspective'. *Research Policy* 43: 278–283.
Hess, D. J. (2019) 'Cooler Coalitions for a Warmer Planet: A Review of Political Strategies for Accelerating Energy Transitions'. *Energy Research & Social Science* 57 (November):101246. https://doi.org/10.1016/j.erss.2019.101246.
Isoaho, K., and K. Karhunmaa (2019) 'A Critical Review of Discursive Approaches in Energy Transitions'. *Energy Policy* 128: 930–942.
Johnstone, P., and P. Newell (2018) 'Sustainability Transitions and the State'. *Environmental Innovations and Societal Transitions* 27: 72–82.
Kanger, L., and J. Schot (2018) Deep transitions: Theorizing the long-term patterns of socio-technical change. *Environmental Innovation and Societal Transitions*.
Kern, F. (2011) 'Ideas, Institutions and Interests: Explaining Policy Divergence in Fostering 'System Innovations' towards Sustainability'. *Environment and Planning C: Government and Policy* 29(6): 1116–1134.
Kumar, A., J. Höffken, and A. Pols (eds). (2021) *Dilemmas of Energy Transitions in the Global South: Balancing Urgency and Justice*. 1st ed. Abingdon: Routledge.
Kamboj, P., and R. Tongia (2018) 'India Railways and Coal: An Unsustainable Interdependency'. New Delhi: Brookings India.
Kuzemko, C., A. Lawrence, and M. Watson (2019) 'New Directions in the International Political Economy of Energy'. *Review of International Political Economy* 26(1): 1–24.
Kuzemko, C., M. Lockwood, C. Mitchell, and R. Hoggett (2016) 'Governing for Sustainable Energy System Change: Politics, Contexts and Contingency'. *Energy Research & Social Science* 12: 96–105.
Lahiri-Dutt, K. (2016) 'The Diverse Worlds of Coal in India: Energising the Nation, Energising Livelihoods'. *Energy Policy* 99: 203–213. https://doi.org/10.1016/j.enpol.2016.05.045.
Lawhon, M., and J. T. Murphy (2012) 'Socio-technical Regimes and Sustainability Transitions: Insights from Political Ecology'. *Progress in Human Geography* 36(3): 354–378.
Lecavalier, E., and C. Harrington (2017) 'Entangling Carbon Lock-in: India's Coal Constituency'. *Crime, Law and Social Change* 68(5): 529–546. https://doi.org/10.1007/s10611-017-9701-7.

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Leipprand, A., and C. Flachsland (2018) 'Regime Destabilization in Energy Transitions: The German Debate on the Future of Coal'. *Energy Research & Social Science*, 40: 190–204.
Lennon, M. (2017) 'Decolonizing Energy: Black Lives Matter and Technoscientific Expertise amid Solar Transitions'. *Energy Research & Social Science* 30: 18–27.
Lockwood, M. (2015) 'The Political Dynamics of Green Transformations: Feedback Effects and Institutional Context'. In Scoones, I. et al. (eds), *The Politics of Green Transformations* (pp. 104–119). Routledge.
Lockwood, M., C. Kuzemko, C. Mitchell, and R. Hoggett (2016) 'Historical Institutionalism and the Politics of Sustainable Energy Transitions: A Research Agenda'. *Environment and Planning C: Government Policy* 35(2): 312–333.
Malm, A. (2016) *Fossil Capital: The Rise of Steam Power and the Roots of Global Warming*. London: Verso.
Mazzucato, M. (2011) *The Entrepreneurial State*. London: Demos.
Meadowcroft, J. (2005) 'Environmental Political Economy, Technological Transitions and the State'. *New Political Economy* 10(4): 479–498.
Meadowcroft, J. (2009) 'What about the Politics? Sustainable Development, Transition Management, and Long Term Energy Transitions'. *Policy Sciences* 42: 323–340.
Meckling, J. (2011) *Carbon Coalitions: Business, Climate Politics, and the Rise of Emissions Trading*. Cambridge, MA: MIT Press.
Meckling, J., N. Kelsey, E. Biber, and J. Zysman (2015) 'Winning Coalitions for Climate Policy'. *Science* 349(6253): 1170–1171.
Malik, A., C. Bertram, J. Despres, J. Emmerling, S. Fujimori, A. Garg, E. Kriegler, et al. (2020) 'Reducing Stranded Assets through Early Action in the Indian Power Sector'. *Environmental Research Letters* 15(9): 094091. https://doi.org/10.1088/1748-9326/ab8033.
Moore, J. W. (2015) *Capitalism in the Web of Life: Ecology and the Accumulation of Capital*. London: Verso.
Nayak, S. (2022) 'Coal Extraction, Dispossession and the "Classes of Labour" in Coalfields of Eastern India'. *The Journal of Peasant Studies* 50(7): 2829–2850. https://doi.org/10.1080/03066150.2022.2145955.
Newell, P. (2018) 'Trasformismo or Transformation? The Global Political Economy of Energy Transitions'. *Review of International Political Economy* 26(1): 25–48.
Newell, P. (2020a) 'The Business of Rapid Transition'. *WIRES Climate Change* 11(6): e670.
Newell, P. (2020b) 'Towards a Global Political Economy of Transitions: A Comment on the Transitions Research Agenda'. *Environmental Innovation and Societal Transitions* 34: 344–345.
Newell, P. (2021a) *Power Shift: The Global Political Economy of Energy Transitions*. Cambridge: Cambridge University Press.
Newell, P. (2021b) 'Race and the Politics of Energy Transitions'. *Energy Research & Social Science* 71: 101839.
Newell, P. (2025) *States of Transition: From Governing the Environment to Transforming Society*. Cambridge: Cambridge University Press.
Newell, P., and M. Paterson (2010) *Climate Capitalism: Global Warming and the Transformation of the Global Economy*. Cambridge: Cambridge University Press.
Newell, P., and A. Bumpus (2012) 'The Global Political Ecology of the CDM'. *Global Environmental Politics* 12: 49–67.
Newell, P., and D. Mulvaney (2013). 'The Political Economy of the "Just Transition"'. *The Geographical Journal*, 179(2), 132–140.
Newell, P., and J. Phillips (2016). 'Neoliberal Energy Transitions in the South: Kenyan Experiences'. *Geoforum* 74: 39–48.
Newell, P., and A. Simms (2020). 'Towards a fossil fuel non-proliferation treaty'. *Climate Policy*, 20(8), 1043–1054.

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Newell, P., and F. Daley (2022) 'Cooking up an Electric Revolution: The Political Economy of e-cooking'. *Energy Research & Social Science* 91, 102730.
Newell, P., J. Phillips, and P. Purohit (2011) 'The Political Economy of Clean Development in India: CDM and Beyond'. *IDS Bulletin* 42(3): 89–96.
Newell, P., F. Daley, O. Mikheeva, and I. Pesa (2023) 'Mind the Gap: The Global Governance of Just Transitions'. *Global Policy*. https://doi.org/10.1111/1758-5899.13236.
Ordonez, J. A., M. Jakob, J. C. Steckel, and H. Ward (2023) 'India's Just Energy Transition: Political Economy Challenges across States and Regions'. *Energy Policy* 179: 113621. https://doi.org/10.1016/j.enpol.2023.113621.
Oskarsson, P., K. B. Nielsen, K. Lahiri-Dutt, and B. Roy (2021) 'India's New Coal Geography: Coastal Transformations, Imported Fuel and State-Business Collaboration in the Transition to More Fossil Fuel Energy'. *Energy Research & Social Science* 73 (March): 101903. https://doi.org/10.1016/j.erss.2020.101903.
Pai, S. (2021) 'Fossil Fuel Phase Outs to Meet Global Climate Targets: Investigating the Spatial and Temporal Dimensions of Just Transitions'. University of British Columbia. https://doi.library.ubc.ca/10.14288/1.0398719.
Power, M., et al. (2016) 'The Political Economy of Energy Transitions in Mozambique and South Africa: The Role of the Rising Powers'. *Energy Research and Social Sciences* 17: 10–19.
Power, M., and J. Kirshner (2018) 'Powering the State: The Political Geographies of Electrification in Mozambique'. *Environment and Planning C: Politics and Space* 37(3): 498–518.
Roberts, C., F. Geels, M. Lockwood, P. Newell, H. Schmitz, B. Turnheim, and A. Jordan (2018) 'The Politics of Accelerating Low-Carbon Transitions: Towards a New Research Agenda'. *Energy Research & Social Science* 44: 304–311.
Roy, B., and A. Schaffartzik (2021) 'Talk Renewables, Walk Coal: The Paradox of India's Energy Transition.' *Ecological Economics* 180 (February): 106871. https://doi.org/10.1016/j.ecolecon.2020.106871.
Schindler, S., I. Alami, and N. Jepson (2022) 'Goodbye Washington Confusion, Hello Wall Street Consensus: Contemporary State Capitalism and the Spatialisation of Industrial Strategy'. *New Political Economy* 28(2): 223–240. https://doi.org/10.1080/13563467.2022.2091534.
Scoones, I., M. Leach, and P. Newell (eds.) (2015) *The Politics of Green Transformations*. London: Routledge.
Sharma, K. R., and P. Bhatia (2022) 'How Just and Democratic Is India's Solar Energy Transition? An Analysis of State Solar Policies in India'. In *Climate Justice in India*, edited by P. Kashwan, 1st ed., 50–74. India: Cambridge University Press. www.cambridge.org/core/books/climate-justice-in-india/9728CA91D37ABECD4816C16BA3198873#fndtn-information.
Shen, W., and M. Power (2017) 'Africa and the Export of China's Clean Energy Revolution'. *Third World Quarterly* 38(3): 678–697.
Shove, E., and G. Walker (2007) 'CAUTION! Transitions Ahead: Politics, Practice, and Sustainable Transition Management'. *Environment and planning A*, *39*(4): 763–770.
Shove, E., M. Pantzar, and M. Watson (2012) *The Dynamics of Social Practice*. London: Sage.
Silvester, B. R., and J. K. Fisker (2023) 'A Relational Approach to the Role of the State in Societal Transitions and Transformations Towards Sustainability'. *Environmental Innovation and Societal Transitions* 47: 100717.
Smith, J. (2000) *Biofuels and the Globalization of Risk: The Biggest Change in North-South Relations since Colonialism*. London: Zed Books.
Sovacool, B. (2016) 'How Long Will It Take? Conceptualizing the Temporal Dynamics of Energy Transitions'. *Energy Research & Social Science* 13: 202–215.

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Sovacool, B. K., and M. C. Brisbois (2019) 'Elite Power in Low-Carbon Transitions: A Critical and Interdisciplinary Review'. *Energy Research & Social Science* 57: 101242.
Stirling, A. (2014) 'Transforming Power: Social Science and the Politics of Energy Choices'. *Energy Research & Social Science* 1(1): 83–95.
Strange, S. (1988) *States and Markets: An Introduction to International Political Economy*. London: Pinter.
Strauss, S., S. Rupp, and T. Love (eds.) (2013) *Cultures of Energy: Power, Practices, Technology*. Walnut Creek, CA: Left Coast Press.
Swilling, M., and E. Annecke (2012) *Just Transitions. Explorations of Sustainability in an Unfair World*. Claremont, South Africa: UCT Press.
Tagotra, N. (2017) 'The Political Economy of Renewable Energy: Prospects and Challenges for the Renewable Energy Sector in India Post-Paris Negotiations'. *India Quarterly: A Journal of International Affairs* 73(1): 99–113. https://doi.org/10.1177/0974928416686584.
Unruh, G. C. (2000) 'Understanding Carbon Lock-In'. *Energy Policy* 28: 817–830.
Van de Graaf, T. (2013) *The Politics and Institutions of Global Energy Governance*. Basingstoke: Palgrave.
Van de Graaf, T., B. Sovacool, F. Kern, and M. Klare (eds.) (2016) *The Palgrave Handbook of the International Political Economy of Energy*. London: Palgrave.
Yenneti, K., and R. Day (2016) 'Distributional Justice in Solar Energy Implementation in India: The Case of Charanka Solar Park'. *Journal of Rural Studies* 46 (August): 35–46. https://doi.org/10.1016/j.jrurstud.2016.05.009.

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